GAZİ UNIVERSITY

FACULTY OF ENGINEERING
DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

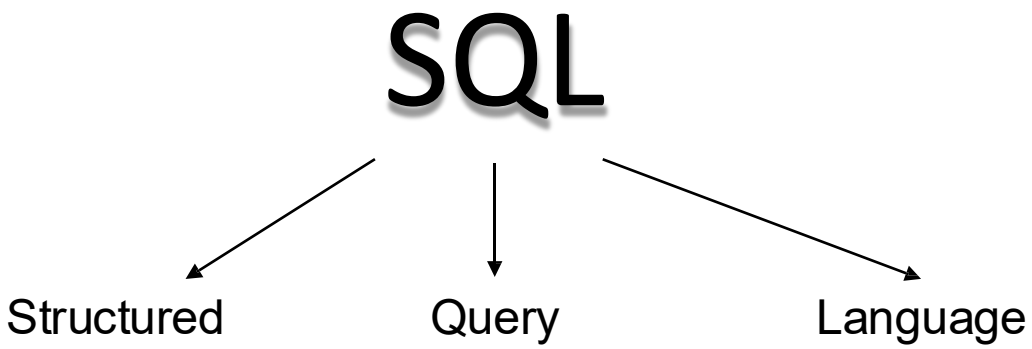
IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 4: LOGICAL DESIGN

Structured Query Language

Intro to SQL and PostgreSQL

Introduction



- SQL is a standard language for storing, processing (inserting, updating, deleting) and retrieving data in databases.
- SQL became the American National Standards Institute (ANSI) standard in 1986 and the International Standards Organization (ISO) standard in 1987.

Introduction

With the SQL in a database, you can:

- Execute a query.
- Retrieve data.
- Add records.
- Update records.
- Delete records.
- Create new tables.
- Create stored procedures.
- Create Functions.
- Create new databases.
- ……

Introduction

- Although SQL is an ANSI/ISO standard, there are different versions of the SQL language.
- To be compatible with the ANSI standard, they all at least similarly support large commands (like SELECT, UPDATE, DELETE, INSERT, WHERE).
- Most SQL database programs have their own proprietary extensions in addition to the SQL standard.

❑ Microsoft SQL Server ➔ Transact SQL, T-SQL

❑ Oracle ➔ Procedural Language extensions to SQL, PL-SQL

❑ PostgreSQL ➔ PL/pgSQL

Introduction

Case Sensitivity in Commands:

- SQL keywords are NOT case sensitive: select and SELECT are the same expressions.

Semicolon After Commands:

- The semicolon is the standard way to separate each SQL statement from the others when multiple SQL statements are executed in the same transaction to the server.
- Semicolons are mandatory in PostgreSQL to terminate most statements and are essential for parsing clear execution boundaries.

Introduction

- Some important command for SQL:

SELECT	Retrieves data from DB	
UPDATE	Updates data in a table.	
DELETE	Deletes data from a table.	
INSERT INTO	Adds data to a table.	
CREATE	**TABLE, DATABASE, INDEX, TRIGGER, STORED PROCEDURE, FUNCTION**	Creates new.
ALTER		Modifies the existing one.
DROP		Deletes the existing one.

Introduction

SQL commands are divided into three basic classes according to the operations they perform:

- Data manipulation/manipulation language (DML, Data Manipulation Language):
  - SELECT, INSERT, UPDATE etc. commands that operate on the data in the table.
- Data definition language (DDL, Data Definition Language):
  - Including the commands used to define the data element starting with CREATE, ALTER, DROP.
- Data control language (DCL, Data Control Language):
  - GRAND, REVOKE etc. These commands for creating user privileges.

Introduction

- Two tier Architecture (Such as Windows Application)

Database Layer

Presentation Layer
(User)

SQL Query

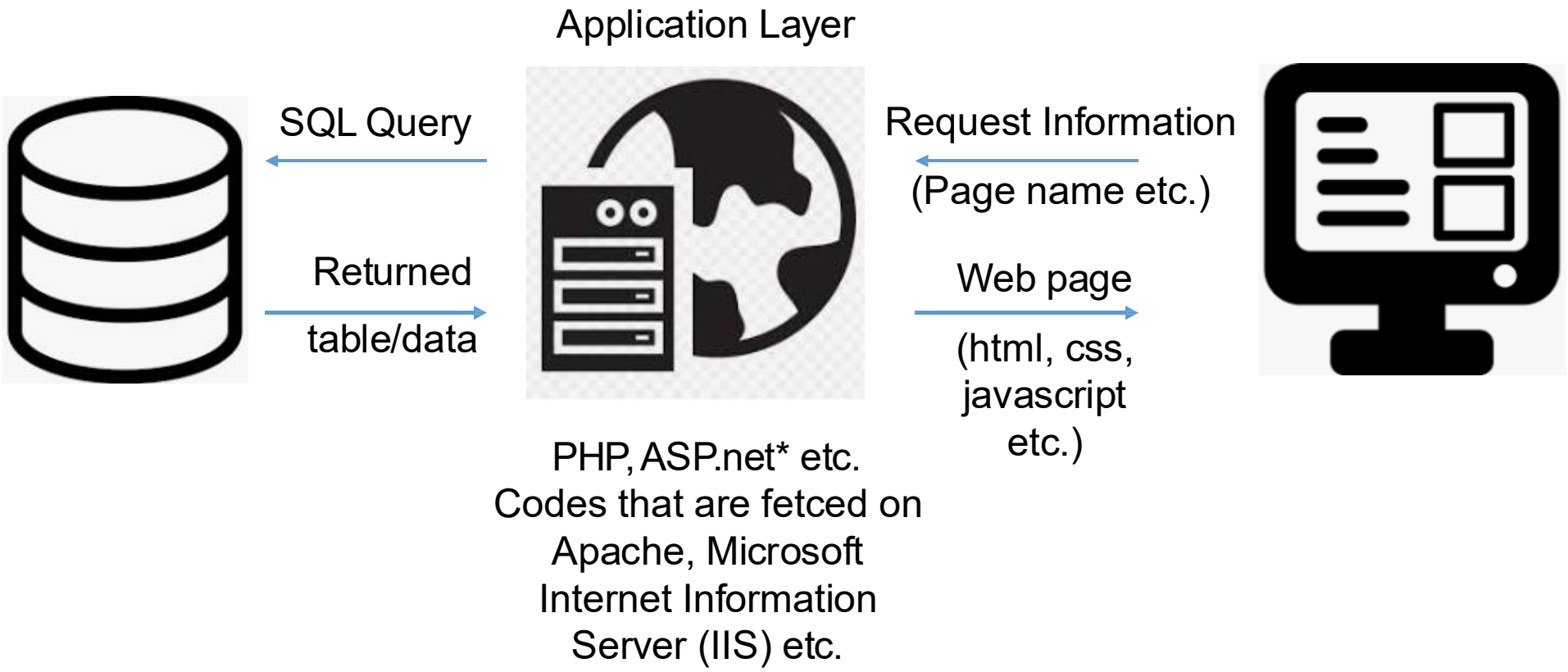
Returned table/data

SQL Server, Oracle, My SQL, Firebird etc.

Interface application coded by Ms Visual Studio, Java vb.

Introduction

- Three tier Architecture (Such as Web Application)

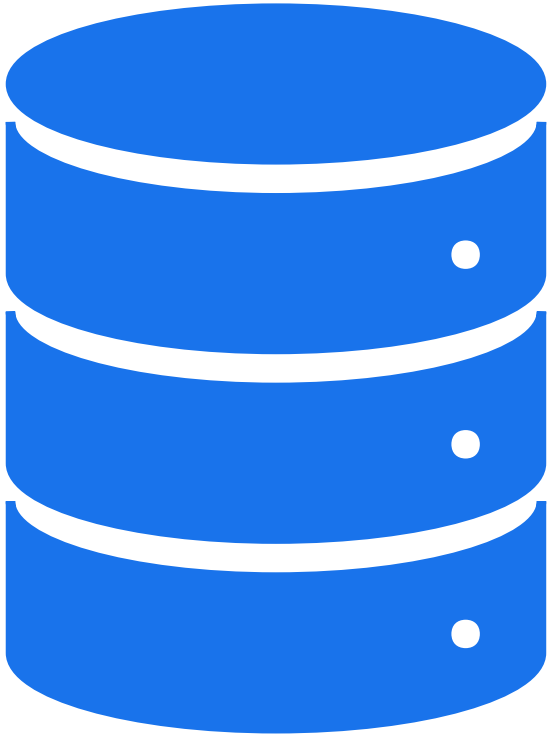


PostgreSQL

The World's Most Advanced Open-Source Relational Database

PostgreSQL presentation

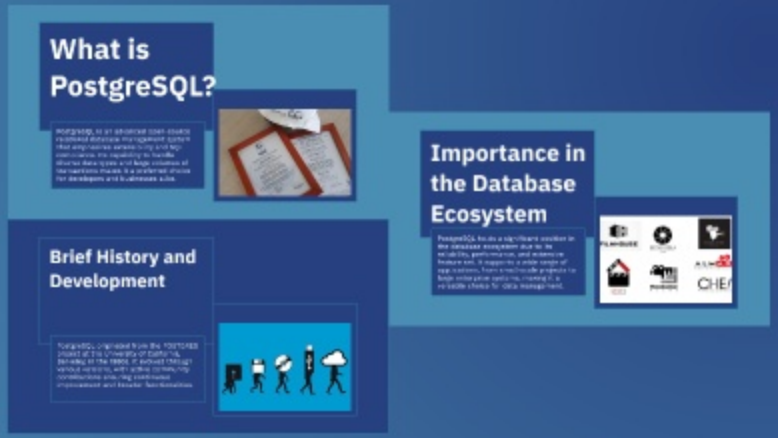
PostgreSQL, often called **Postgres**, is a **powerful, open-source, object-relational database management system (ORDBMS)** used to store, manage, and query structured data using SQL.



PostgreSQL Presentation

An Overview of Core Features, Introduction, and Future Prospects

Introduction to PostgreSQL

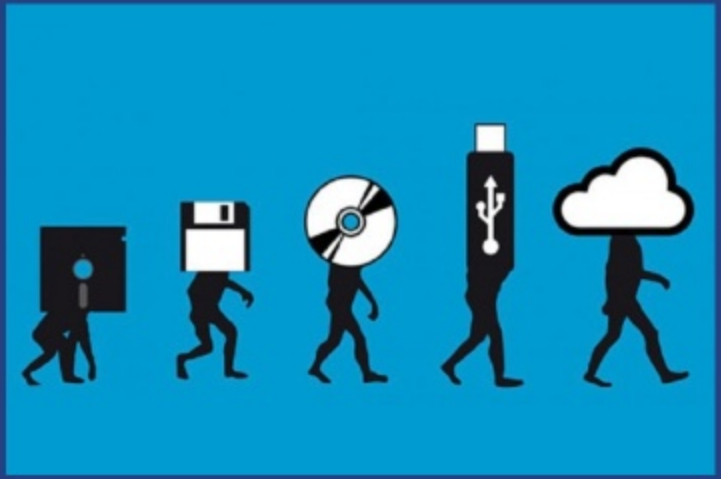


What is PostgreSQL?

PostgreSQL is an advanced open-source relational database management system that emphasizes extensibility and SQL compliance. Its capability to handle diverse data types and large volumes of transactions makes it a preferred choice for developers and businesses alike.

Brief History and Development

PostgreSQL originated from the POSTGRES project at the University of California, Berkeley, in the 1980s. It evolved through various versions, with active community contributions ensuring continuous improvement and broader functionalities.



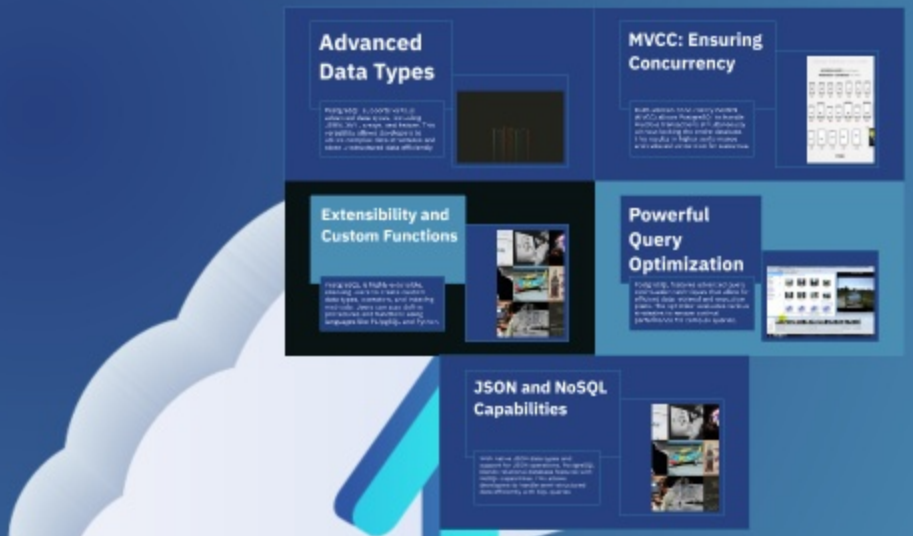
Importance in the Database Ecosystem

PostgreSQL holds a significant position in the database ecosystem due to its reliability, performance, and extensive feature set. It supports a wide range of applications, from small-scale projects to large enterprise systems, making it a versatile choice for data management.

PostgreSQL Presentation

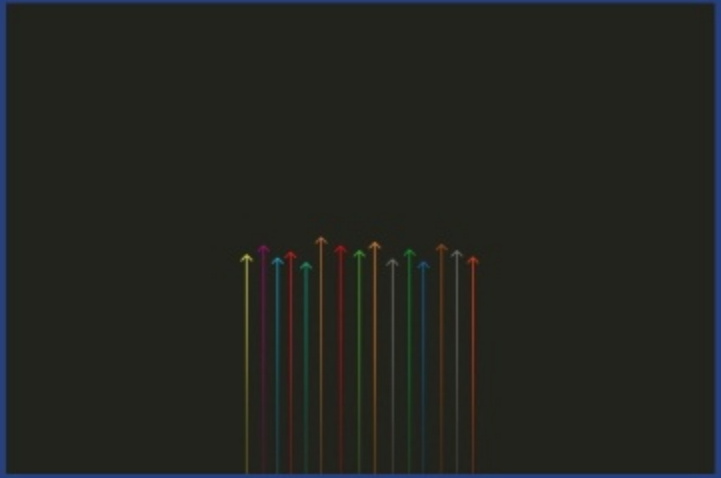
An Overview of Core Features, Introduction, and Future Prospects

Core Features of PostgreSQL



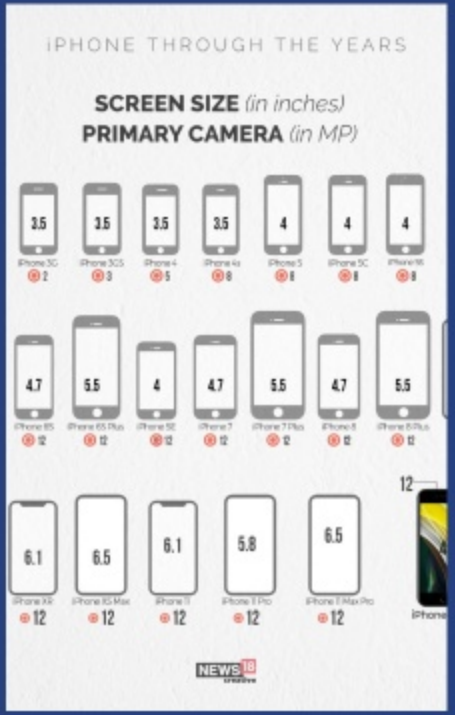
Advanced Data Types

PostgreSQL supports various advanced data types, including JSON, XML, arrays, and hstore. This versatility allows developers to utilize complex data structures and store unstructured data efficiently.



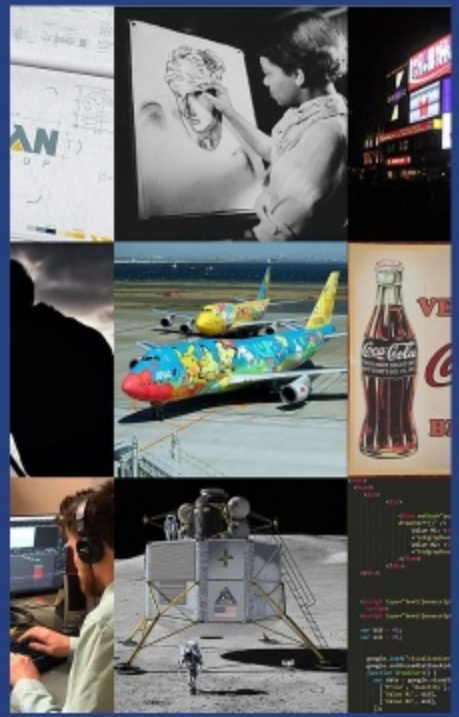
MVCC: Ensuring Concurrency

Multi-Version Concurrency Control (MVCC) allows PostgreSQL to handle multiple transactions simultaneously without locking the entire database. This results in higher performance and reduced contention for resources.



Extensibility and Custom Functions

PostgreSQL is highly extensible, enabling users to create custom data types, operators, and indexing methods. Users can also define procedures and functions using languages like PL/pgSQL and Python.

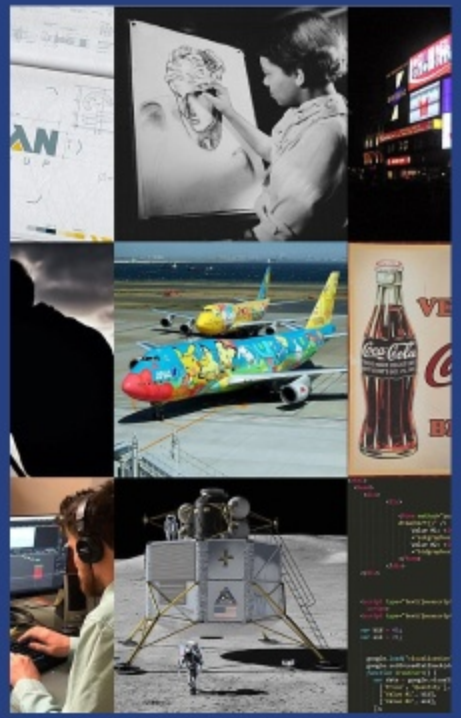


Powerful Query Optimization

PostgreSQL features advanced query optimization techniques that allow for efficient data retrieval and execution plans. The optimizer evaluates various strategies to ensure optimal performance for complex queries.

JSON and NoSQL Capabilities

With native JSON data types and support for JSON operations, PostgreSQL blends relational database features with NoSQL capabilities. This allows developers to handle semi-structured data efficiently with SQL queries.



PostgreSQL Presentation

An Overview of Core Features, Introduction, and Future Prospects

Installation and Configuration

System Requirements

PostgreSQL requires a 64-bit processor and a minimum of 1 GB RAM for basic operations, with 2 GB recommended for performance. It supports various operating systems, like Linux, Windows, and macOS, ensuring flexibility in deployment environments.

Installation Steps

To install PostgreSQL, download the appropriate binary package from the official website, followed by running the installer. For Linux users, package managers can be employed to simplify the installation process, ensuring all dependencies are addressed automatically.

Initial Configuration Settings

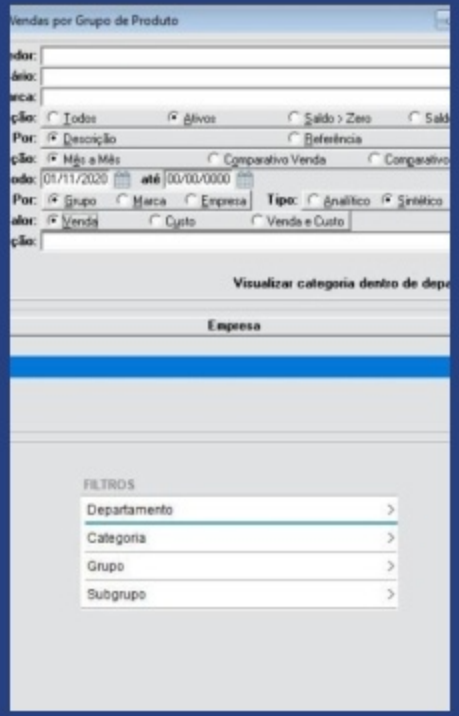
PostgreSQL requires configuring essential settings such as port number and data directory during the initial setup. It's also advisable to set up authentication methods and set up roles for user access management for enhanced security and functionality.

System Requirements

PostgreSQL requires a 64-bit processor and a minimum of 1 GB RAM for basic operations, with 2 GB recommended for performance. It supports various operating systems like Linux, Windows, and macOS, ensuring flexibility in deployment environments.

Installation Steps

To install PostgreSQL, download the appropriate binary package from the official website, followed by running the installer. For Linux users, a package manager can be employed to simplify the installation process, ensuring all dependencies are addressed automatically.



Initial Configuration Settings

PostgreSQL requires configuring essential settings such as port number and data directory during its initial setup. It's also advisable to adjust authentication methods and set up roles for user access management for enhanced security and functionality.

PostgreSQL Presentation

An Overview of Core Features, Introduction, and Future Prospects

Use Cases and Applications of PostgreSQL

Web Development

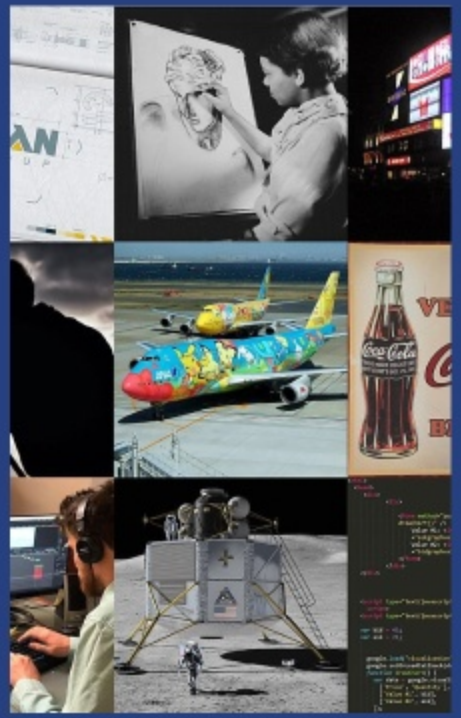
PostgreSQL supports various web frameworks such as Django and Ruby on Rails, allowing developers to create dynamic websites. Its ability to handle JSON and traditional relational data makes it ideal for modern web applications that require flexibility and performance.

Data Warehousing

PostgreSQL is effective for data warehousing due to its ability to handle large volumes of data efficiently. With features such as partitioning and indexing, it allows for quick data retrieval and analysis, essential for decision-making processes.

Geospatial Applications

PostgreSQL, with its PostGIS extension, provides powerful geospatial capabilities for handling location-based data. This is particularly useful for applications in urban planning, environmental monitoring, and navigation systems.



Business Intelligence Tools

PostgreSQL serves as a back-end database for various Business Intelligence (BI) tools like Tableau and Power BI, enabling effective data visualization and reporting. It supports complex queries and aggregations crucial for insightful business analysis.

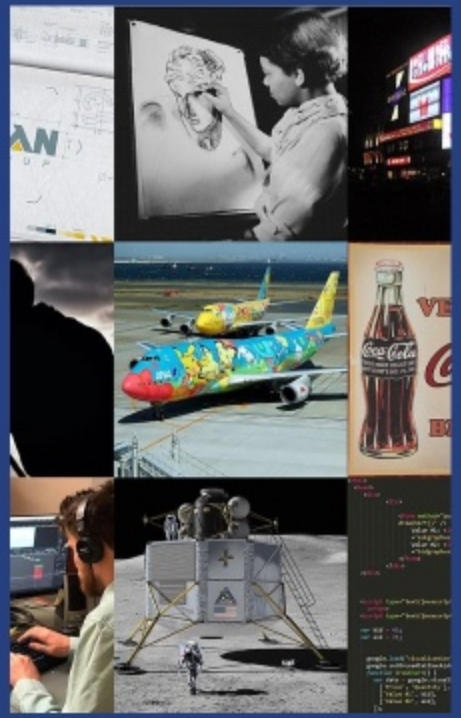
PostgreSQL Presentation

An Overview of Core Features, Introduction, and Future Prospects

Future of PostgreSQL

Upcoming Features and Enhancements

PostgreSQL's roadmap includes significant features like native support for stored procedures in various languages, improved partitioning, and enhanced performance optimizations for cloud environments. These enhancements aim to increase efficiency and usability, ensuring PostgreSQL remains competitive in the database market.

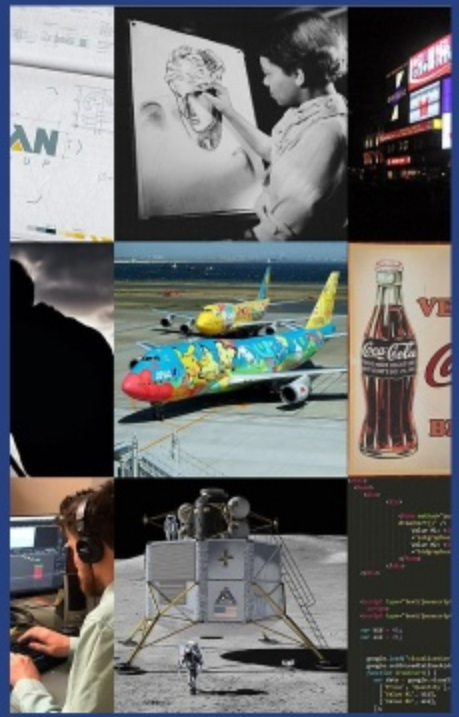


Community and Support

The PostgreSQL community is vibrant and expansive, providing extensive documentation, forums, and mailing lists for support. Regular community-driven updates ensure that PostgreSQL benefits from feedback and contributions from users worldwide, fostering continuous improvement and innovation.

Trends in Database Technologies

As database technologies evolve, PostgreSQL adapts by integrating with cloud solutions, enhancing scalability, and supporting distributed databases. Its focus on performance and flexibility positions PostgreSQL as a primary choice for modern application requirements amidst the rise of NoSQL and NewSQL systems.



PostgreSQL Presentation

An Overview of Core Features, Introduction, and Future Prospects

THE END

CHECK OUT COURSE WEBSITE: HTTPS://GAZI-END-MUH.GITHUB.IO

GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM